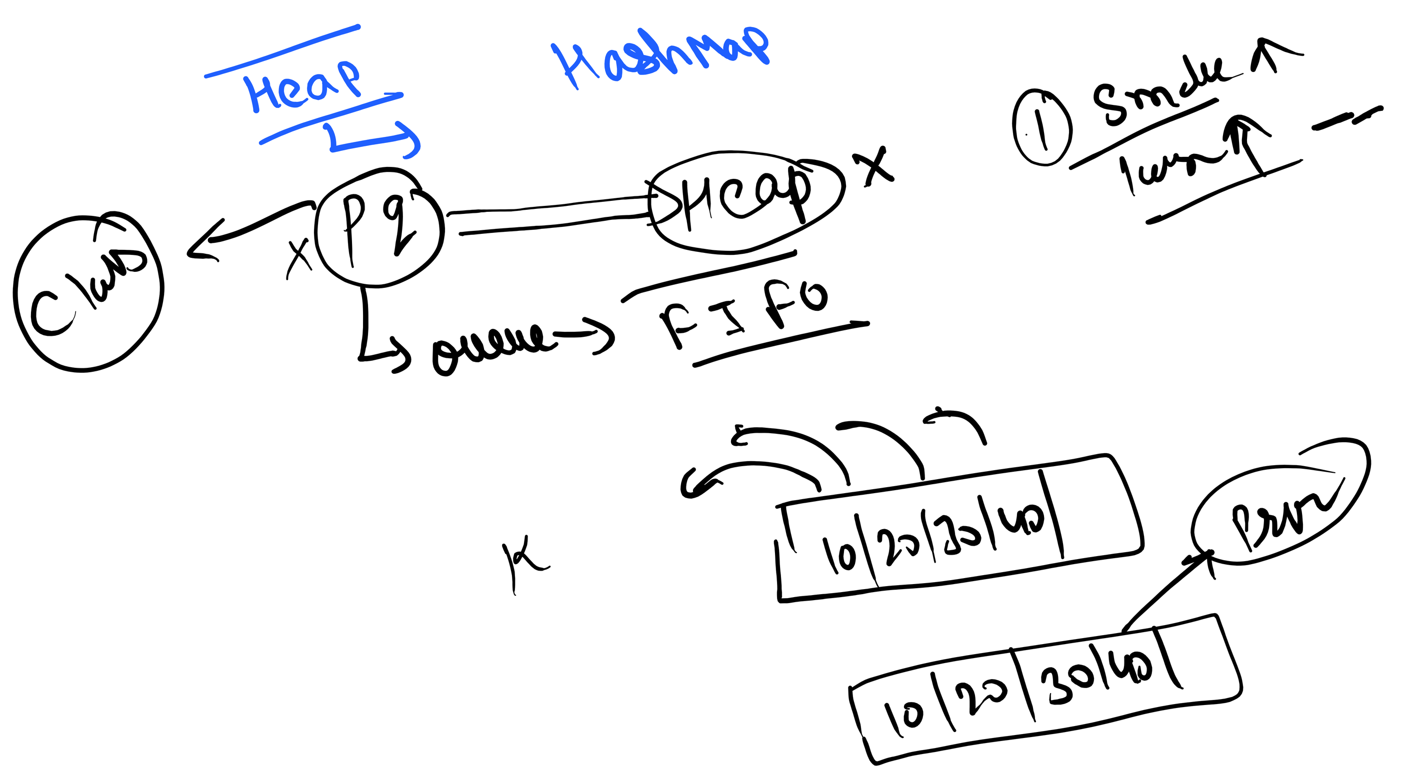
Lec-34

17/05/26 4:53 PM

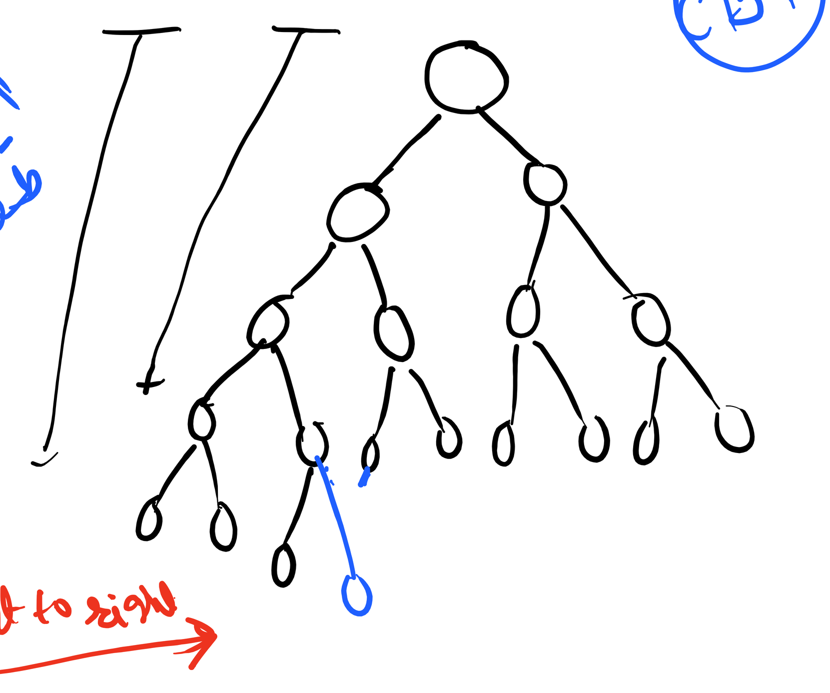


It is complete Binary tree

CBT

smaller ↑ min heap

larger value ↑ max heap



parent<child

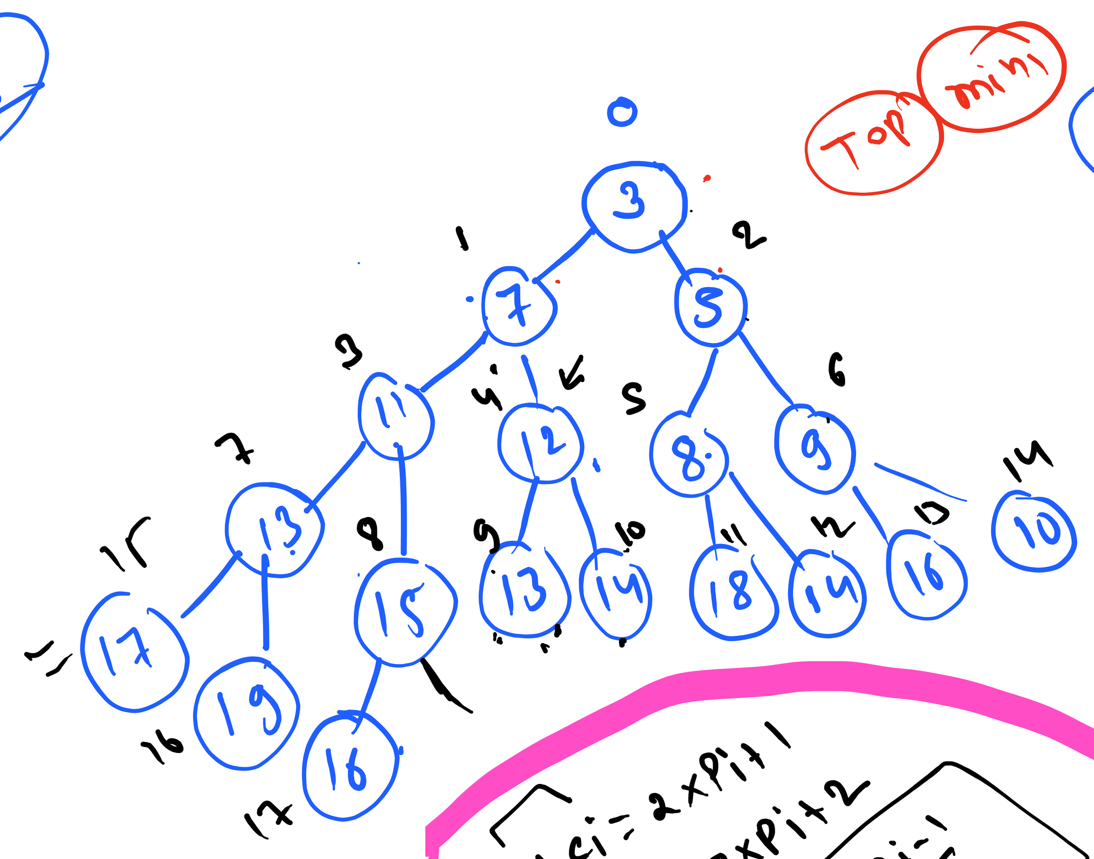
Top mini

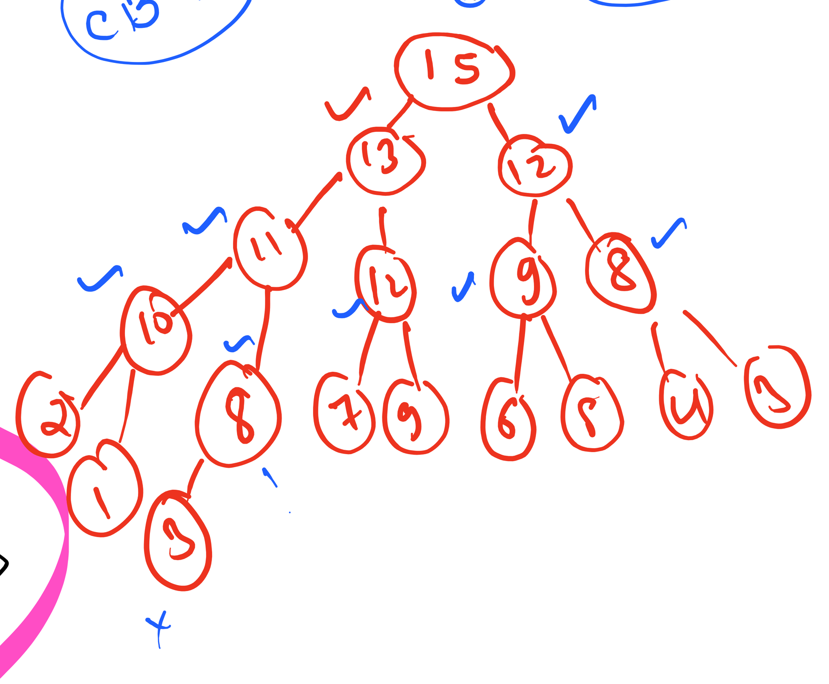
min heap

CBT ✓

max

max-heap





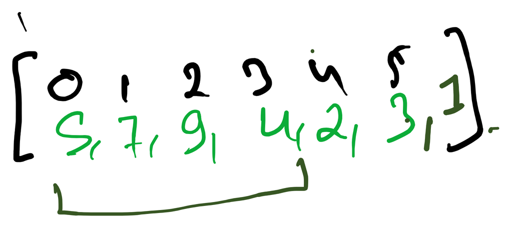
$lci = 2 \times pi + 1$

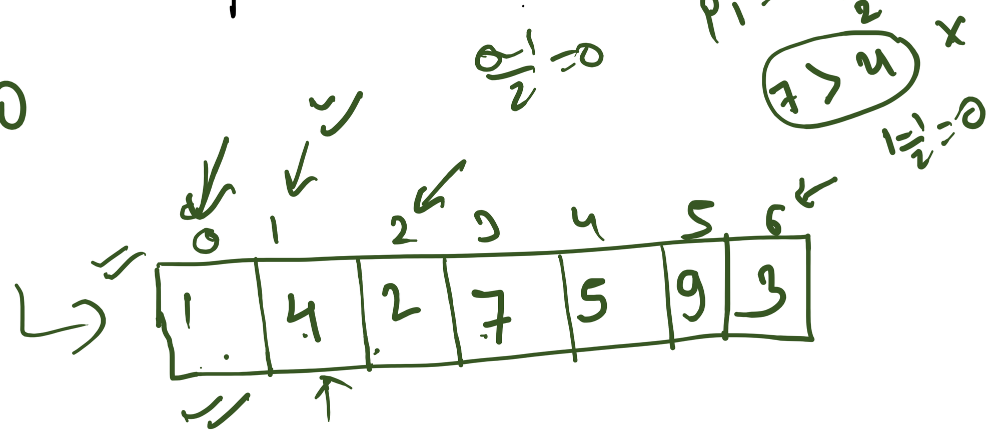
$rci = 2 \times pi + 2$

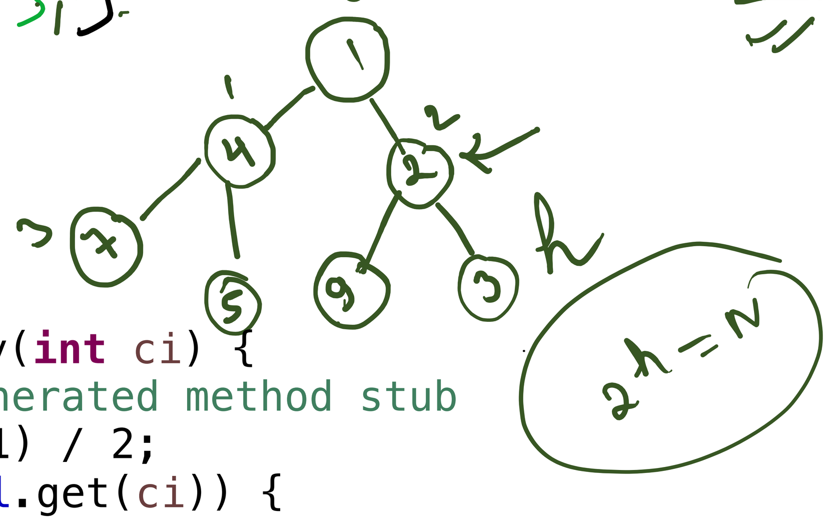
$pi = \frac{ci - 1}{2}$

	Array Sorted	Array unsorted	Heap
Add	O(n)	O(1)	log(n)
remove min	O(n)	O(n)	log(n)
get min	O(1)	O(n)	O(1)

min heap







$\frac{0-1}{2} = 0$

$pi = \frac{3-1}{2} = 1$

7 > 4 ×

$\frac{1-1}{2} = 0$

$\frac{2-1}{2} = 0$

$\frac{4-1}{2} = 1$

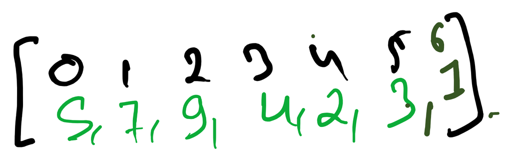
$\frac{1-1}{2} = 0$

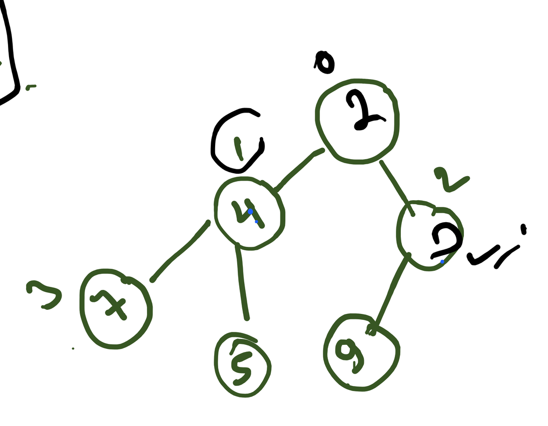
$\frac{5-1}{2} = 2$

$\frac{6-1}{2} = 2$

$2^h = N$

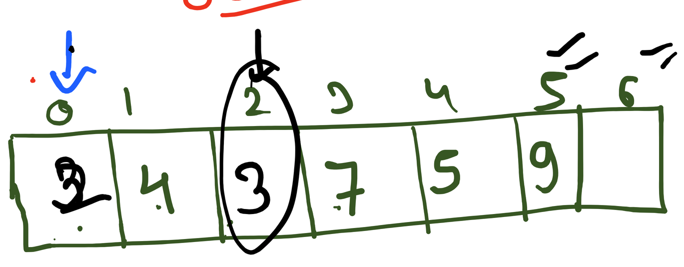
```
private void upheapify(int ci) {
	// TODO Auto-generated method stub
	int pi = (ci - 1) / 2;
	if(ll.get(pi)>ll.get(ci)) {
		swap(pi,ci);
	}
}
```





O(n)

log(n)



```
private void downheapify(int pi) {
	// TODO Auto-generated method stub
	int lci = 2 * pi + 1;
	int rci = 2 * pi + 2;
	int mini = pi;
	if (ll.get(lci) < ll.get(mini)) {
		mini = lci;
	}
	if (ll.get(rci) < ll.get(mini)) {
		mini = rci;
	}
	if(mini!=pi) {
		swap(pi, mini);
		dw-C min)
	}
}
```

$2 \times 2 + 1 = 5$

$2 \times 2 + 1 = 6$